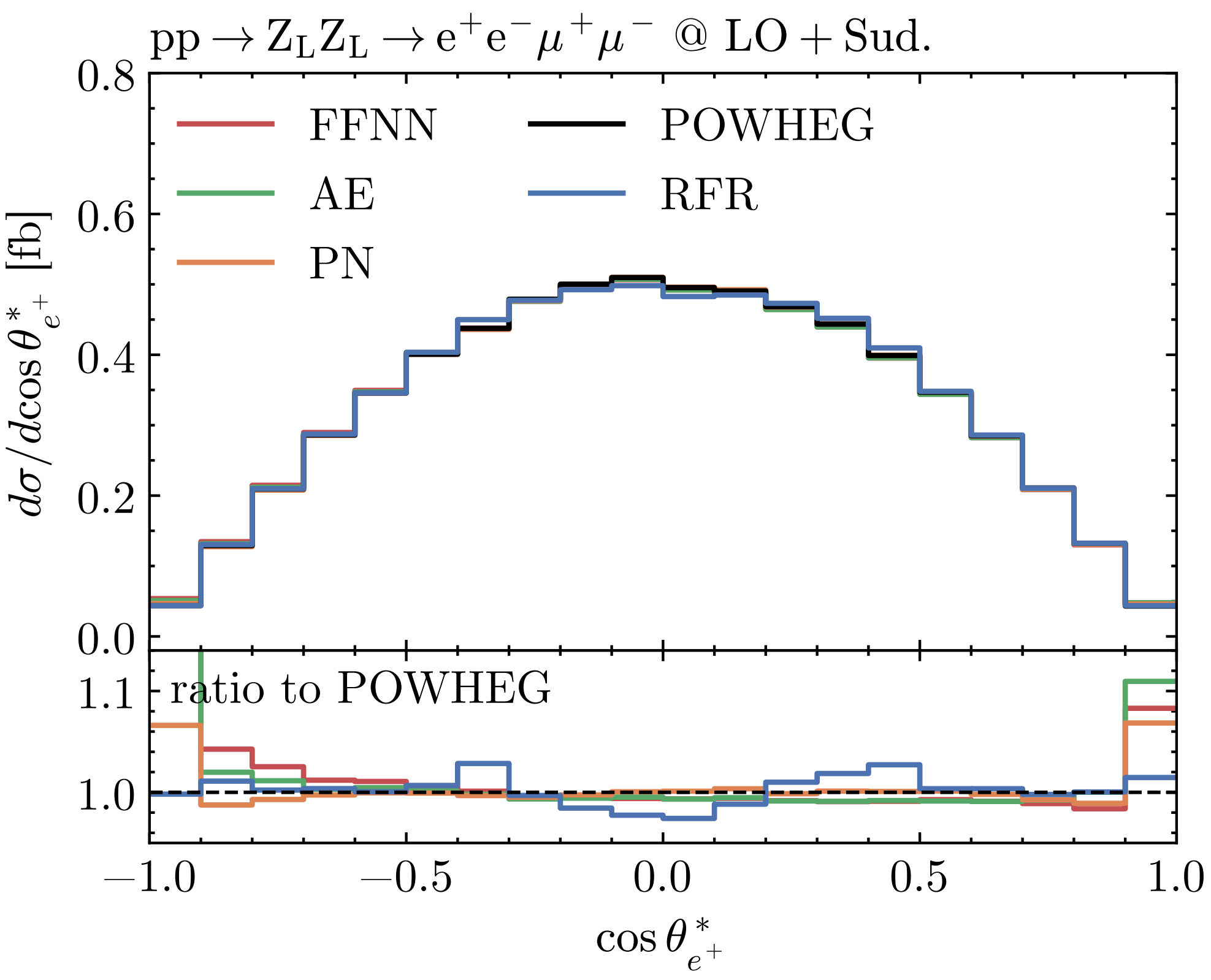
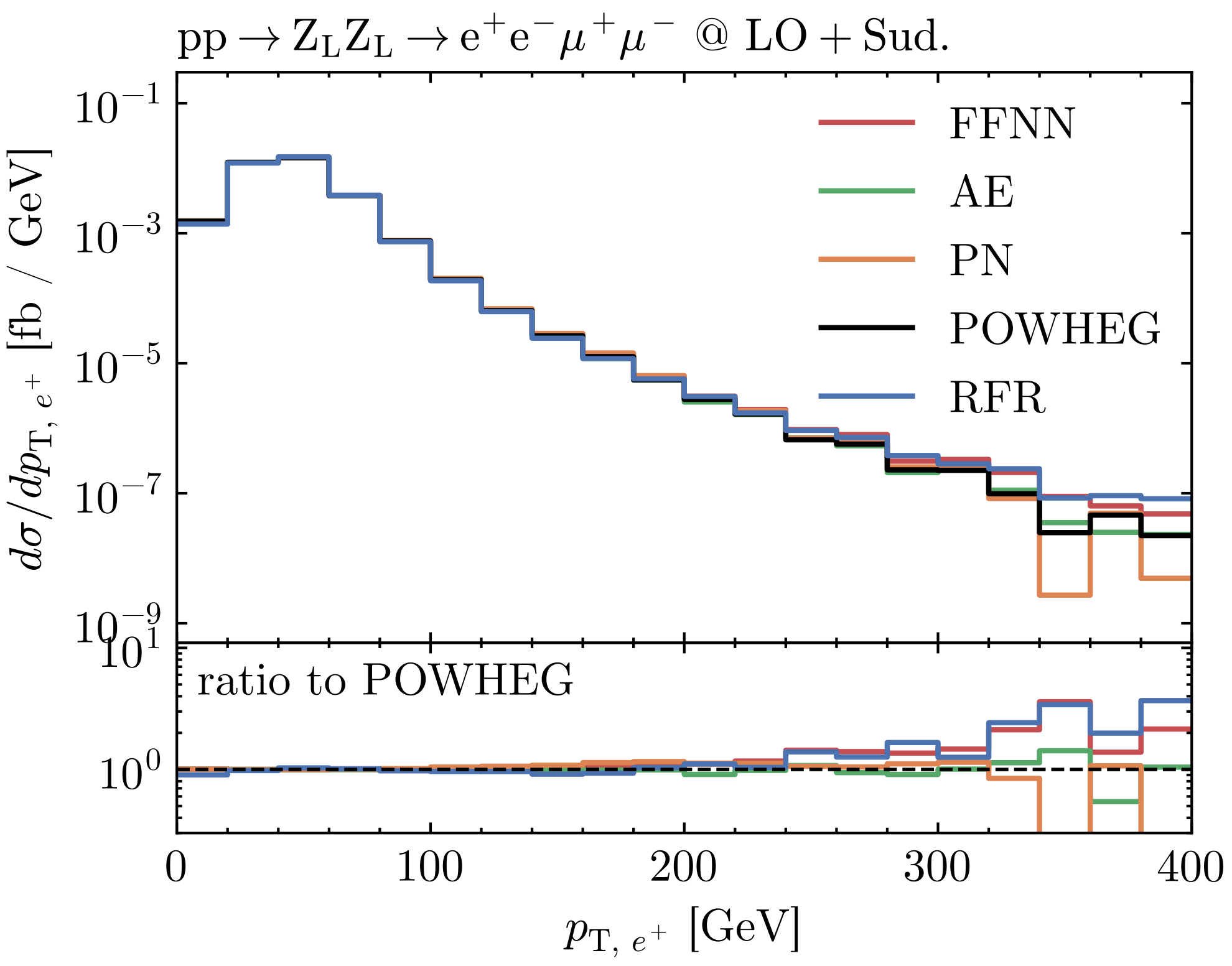
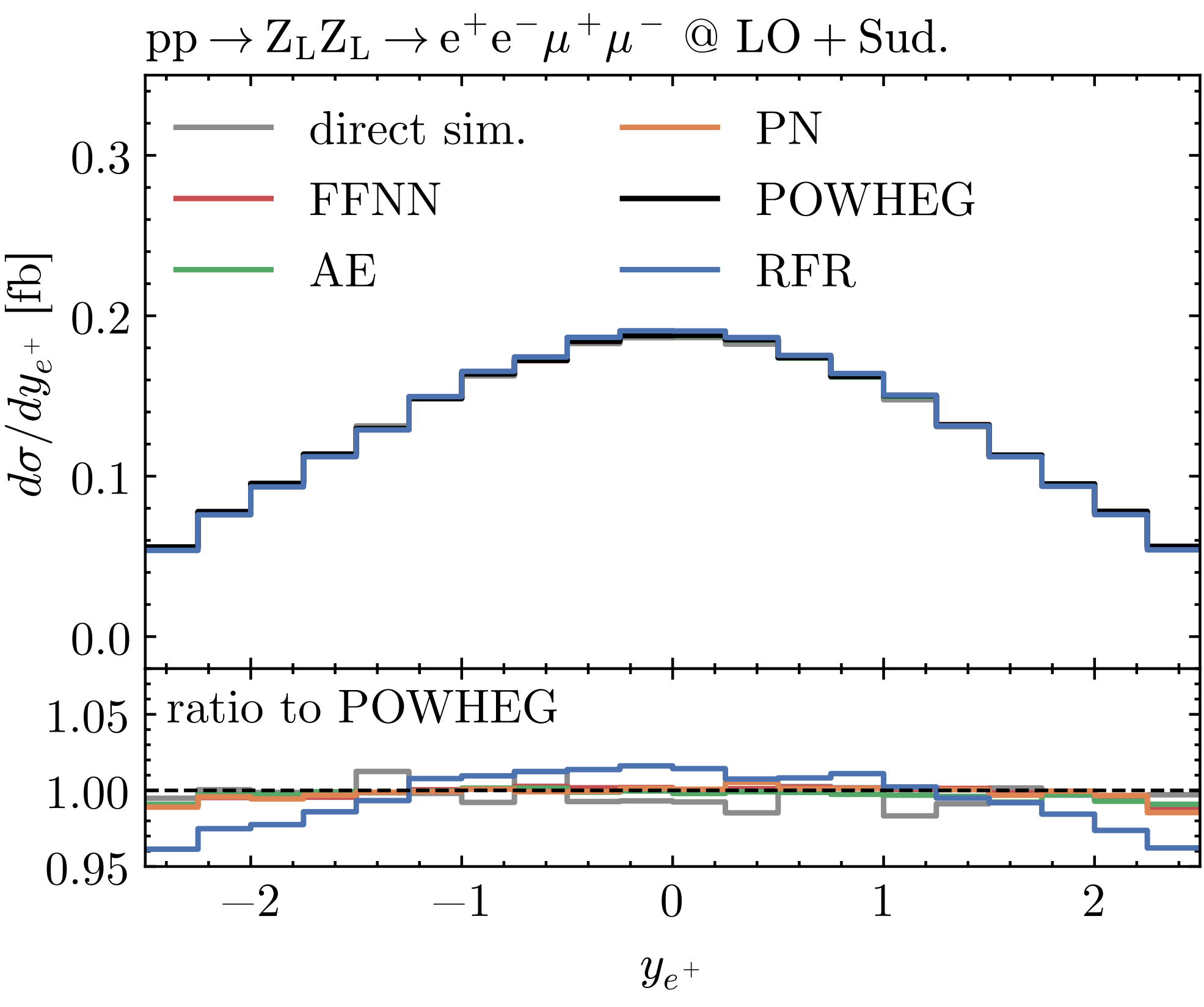
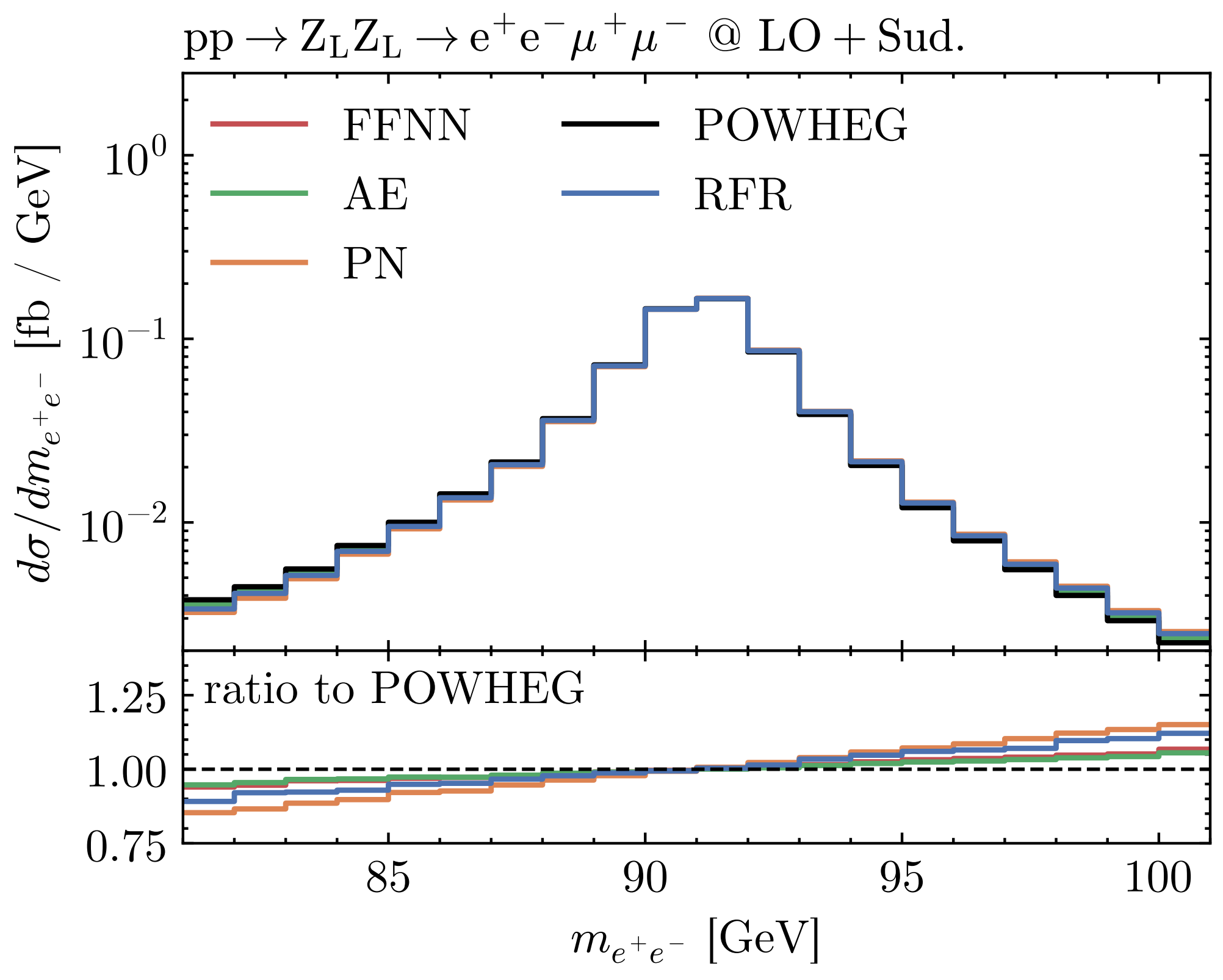
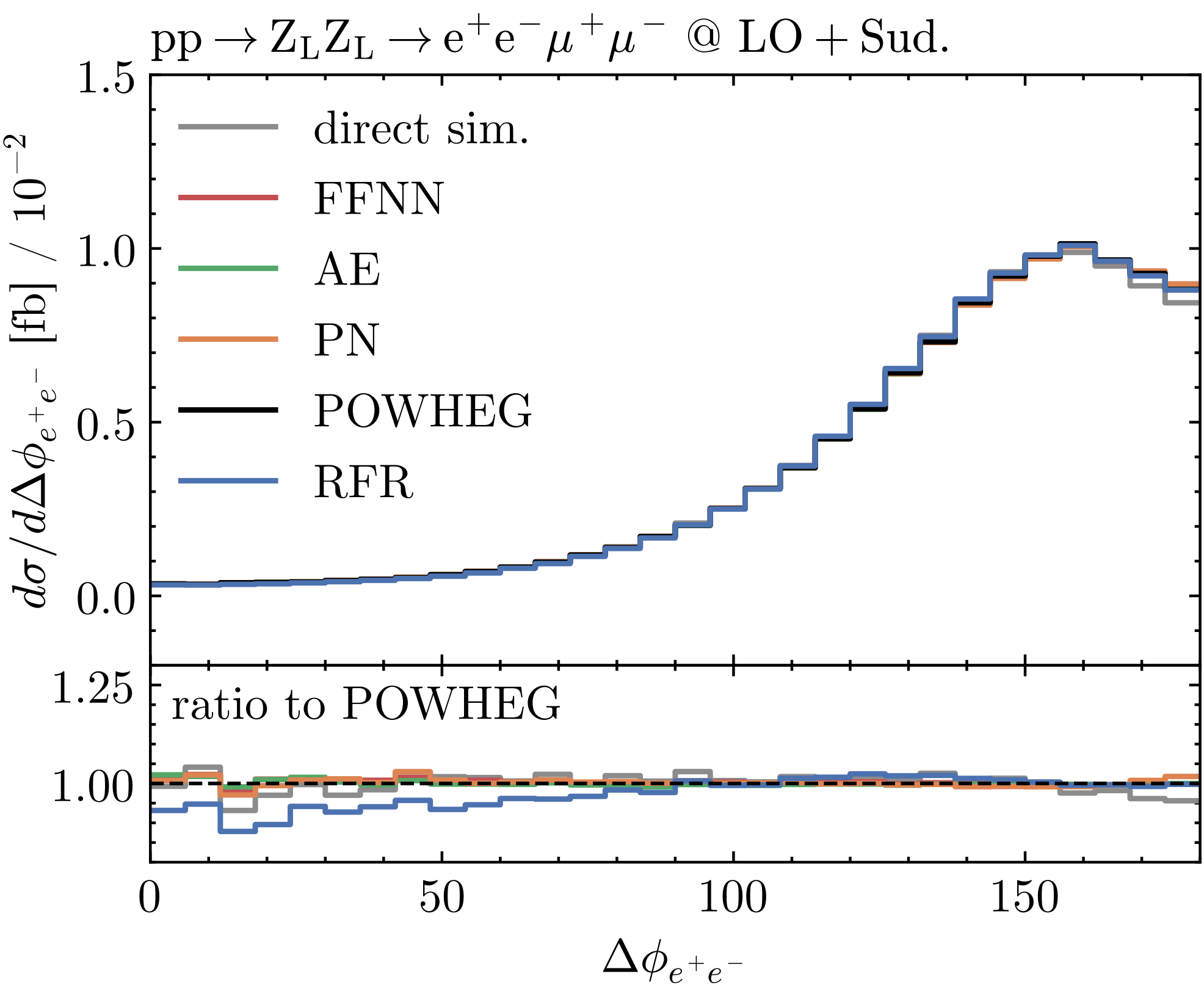
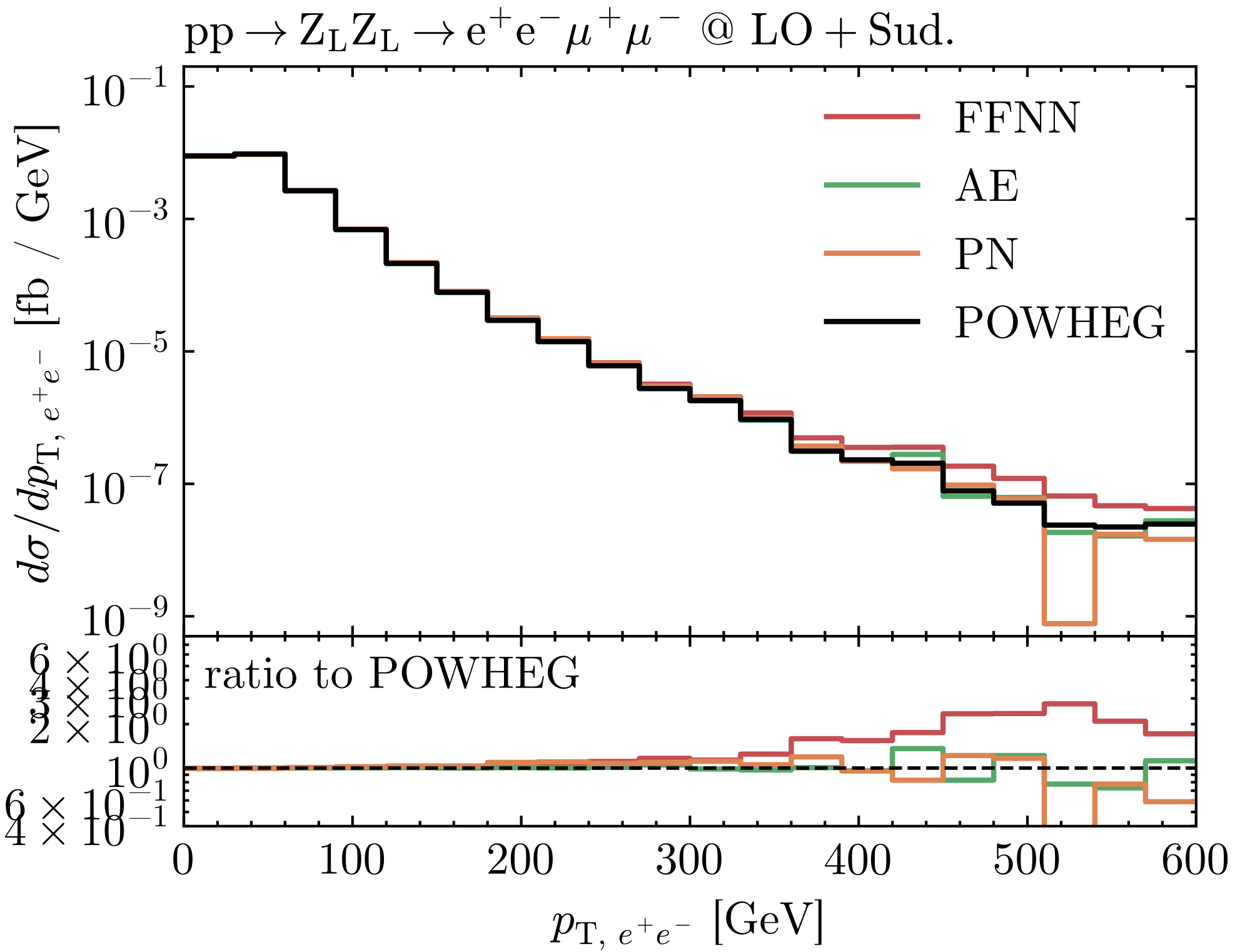

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO + Sud.
FFNN
AE
PN
POWHEG
RFR
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

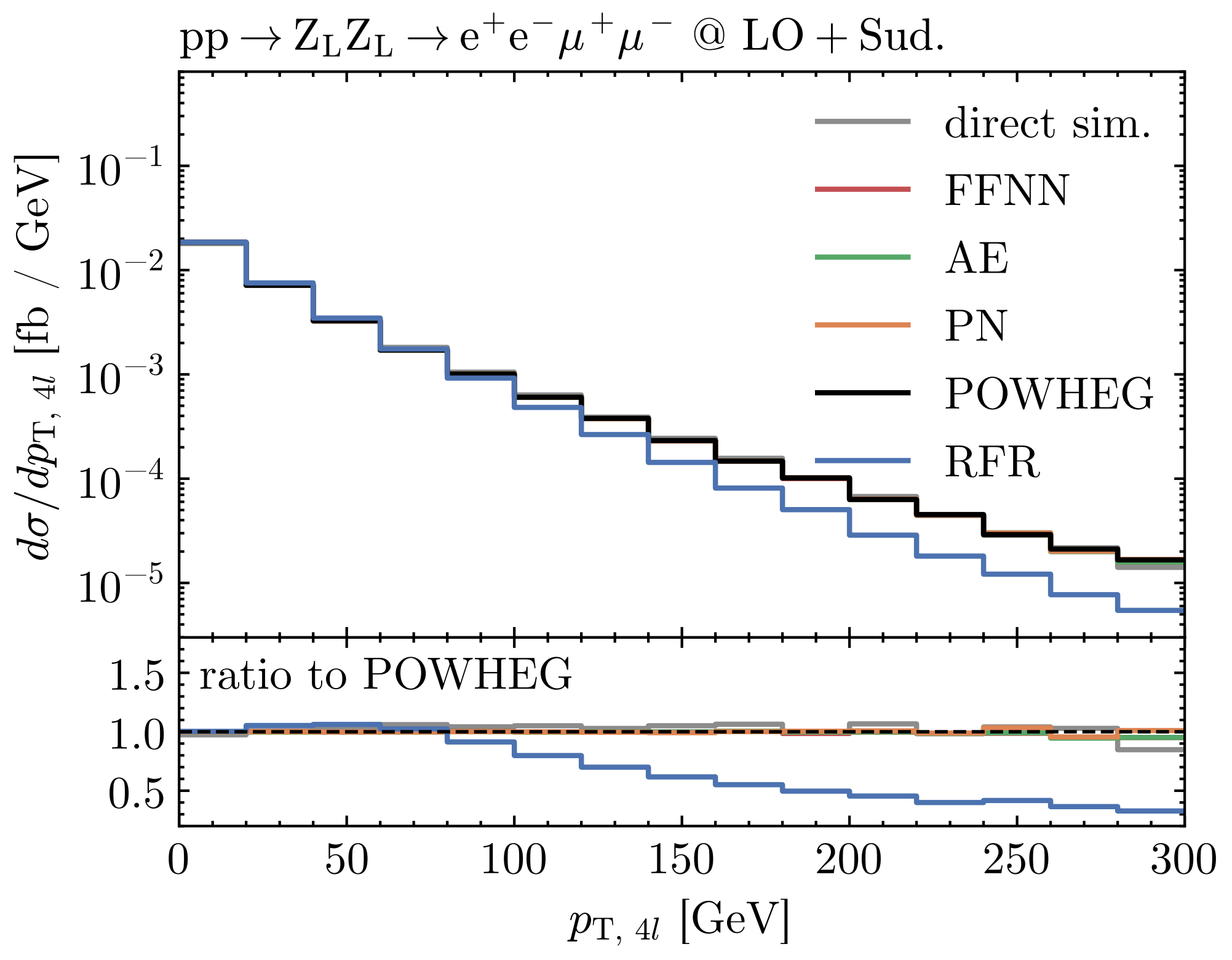

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO + Sud.
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dp_{\mathrm{T},\,e^+}$ [fb / GeV]
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
ratio to POWHEG
10^{1}
10^{0}
0
100
200
300
400
$p_{\mathrm{T},\,e^+}$ [GeV]

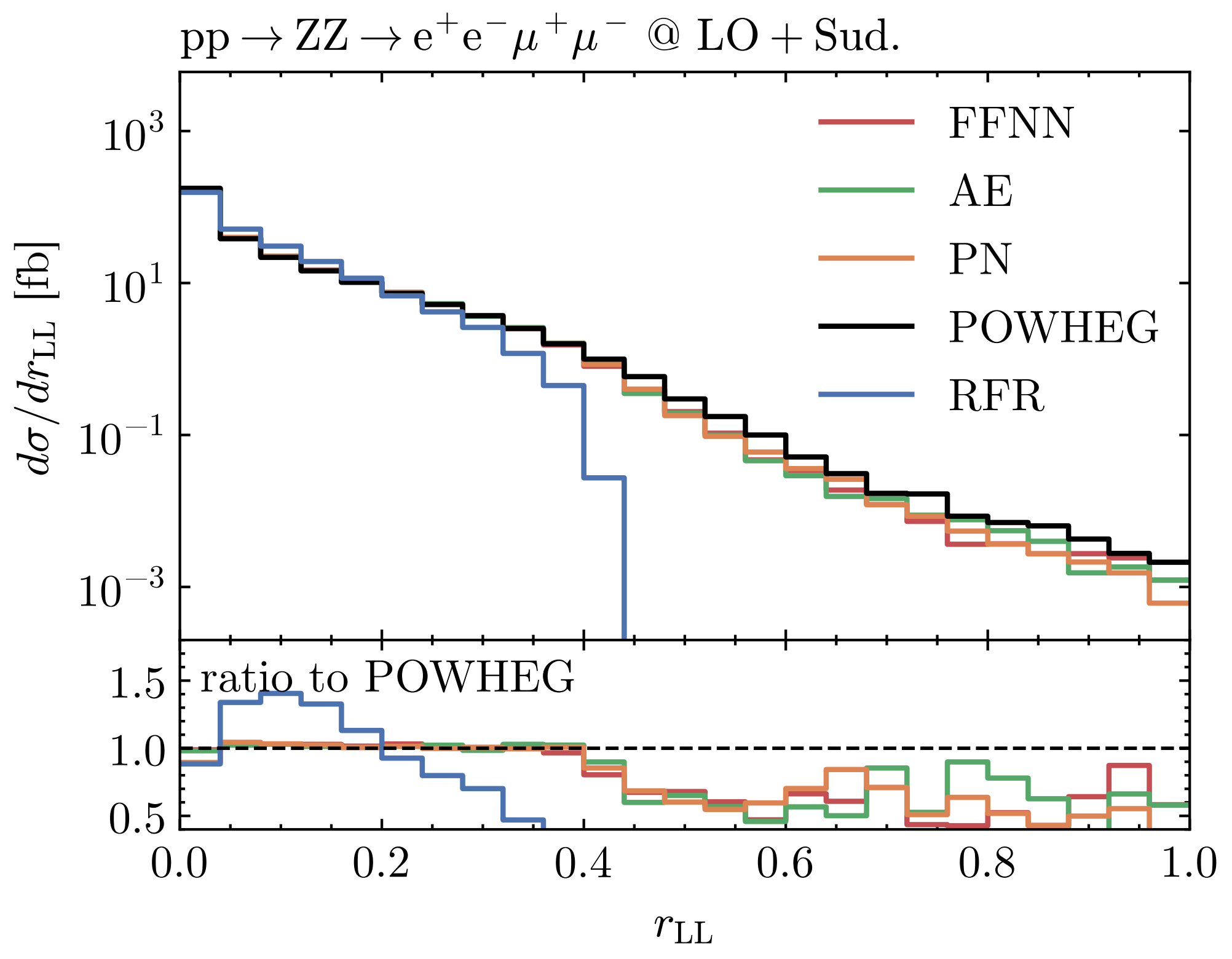

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO + Sud.
direct sim.
FFNN
AE
PN
POWHEG
RFR
$d\sigma / dy_{e^+}$ [fb]
0.0
0.1
0.2
0.3
ratio to POWHEG
0.95
1.00
1.05
−2
−1
0
1
2
y_{e^+}


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO + Sud.
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dm_{e^+e^-}$ [fb / GeV]
10^0
10^{-1}
10^{-2}
ratio to POWHEG
1.25
1.00
0.75
85
90
95
100
$m_{e^+e^-}$ [GeV]


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO + Sud.
direct sim.
FFNN
AE
PN
POWHEG
RFR
$d\sigma / d\Delta\phi_{e^+e^-}$ [fb] / 10^{-2}
1.5
1.0
0.5
0.0
ratio to POWHEG
1.25
1.00
0
50
100
150
$\Delta\phi_{e^+e^-}$


pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻ @ LO + Sud.
FFNN
AE
PN
POWHEG
dσ/dp_T, e⁺e⁻ [fb / GeV]
ratio to POWHEG
p_T, e⁺e⁻ [GeV]
0
100
200
300
400
500
600


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO + Sud.
direct sim.
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dp_{\mathrm{T},\,4l}$ [fb / GeV]
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
ratio to POWHEG
1.5
1.0
0.5
0
50
100
150
200
250
300
$p_{\mathrm{T},\,4l}$ [GeV]


pp → ZZ → e+e−μ+μ− @ LO + Sud.
FFNN
AE
PN
POWHEG
RFR
dσ/dr_LL [fb]
10^3
10^1
10^−1
10^−3
ratio to POWHEG
1.5
1.0
0.5
0.0
0.2
0.4
0.6
0.8
1.0
r_LL